INCH-POUND

MIL-DTL-17/128B
w/AMENDMENT 2
25 February 2005
SUPERSEDING
MIL-C-17/128B
AMENDMENT 1
18 July 1985

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/128-RG400

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-17.

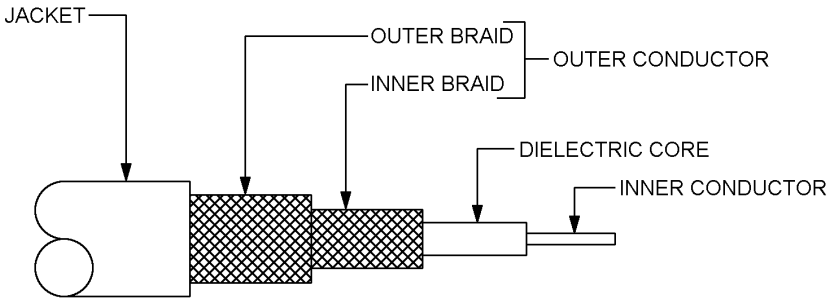


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Nineteen strands of silver-coated copper wire at .008 inch each. Overall diameter: .0384 inch ± .0010.
Dielectric core	Type F-1: Solid extruded PTFE. Diameter: .116 inch ± .005.
Outer conductor	Double braid of AWG size 36 silver-coated copper wire. Diameter: .171 inch maximum.
Inner braid	Coverage: 94.8% nominal Carriers: 16 Ends: 7 Picks/inch: 11.5 ± 10%
Outer braid	Coverage: 93.6% nominal Carriers: 16 Ends: 7 Picks/inch: 14.5 ± 10%
Jacket	Type IX: Diameter: .195 inch ± .005.

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ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 12.4 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -55°C to + 200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.91 ohm per 100 feet.

Elongation: 10 percent, minimum.

Tensile strength: Not applicable.

Engineering notes: This cable is useful in general purpose, high temperature applications (see connector series “TNC”, “BNC”, and “SMA” in accordance with MIL-PRF-39012).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.

Aging stability: Not applicable.

Stress crack resistance: +230° ± 5°C; mandrel size seven times the jacket diameter.

Outer conductor integrity: Not applicable.

Cold bend: -55° C ±2° C.

Dimensional stability: +200°C ± 5°C.

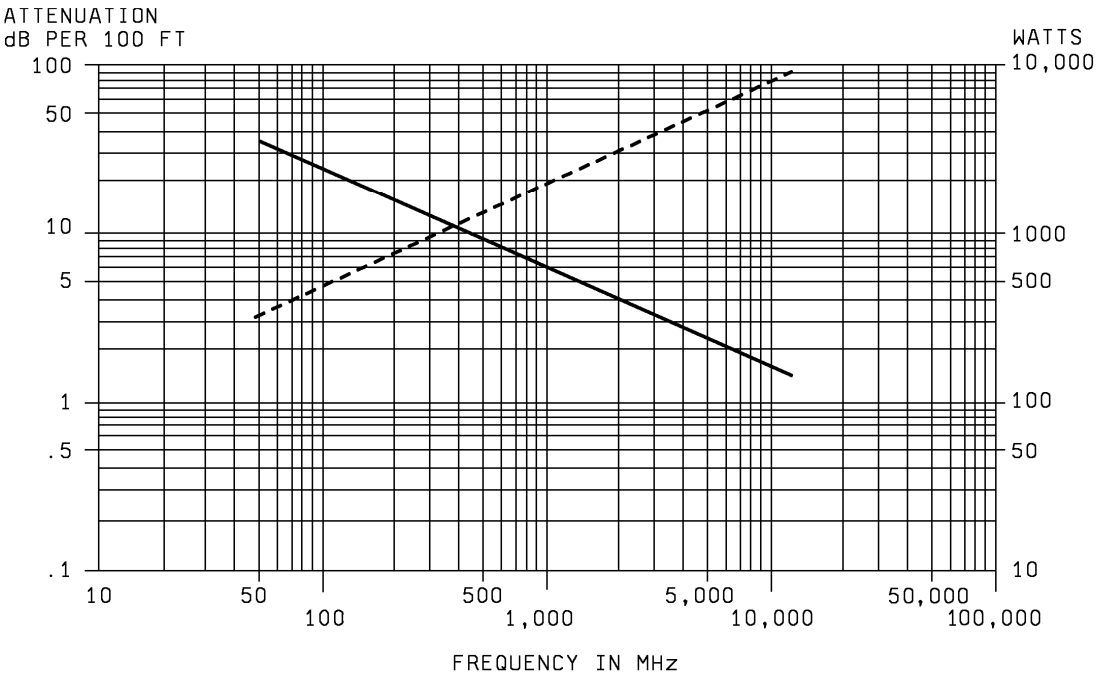
Inner conductor from core: .187 inch, maximum.

Inner conductor from jacket: .312 inch, maximum.

Bendability: Not applicable.

Weight: 5 pounds per 100 feet, maximum.

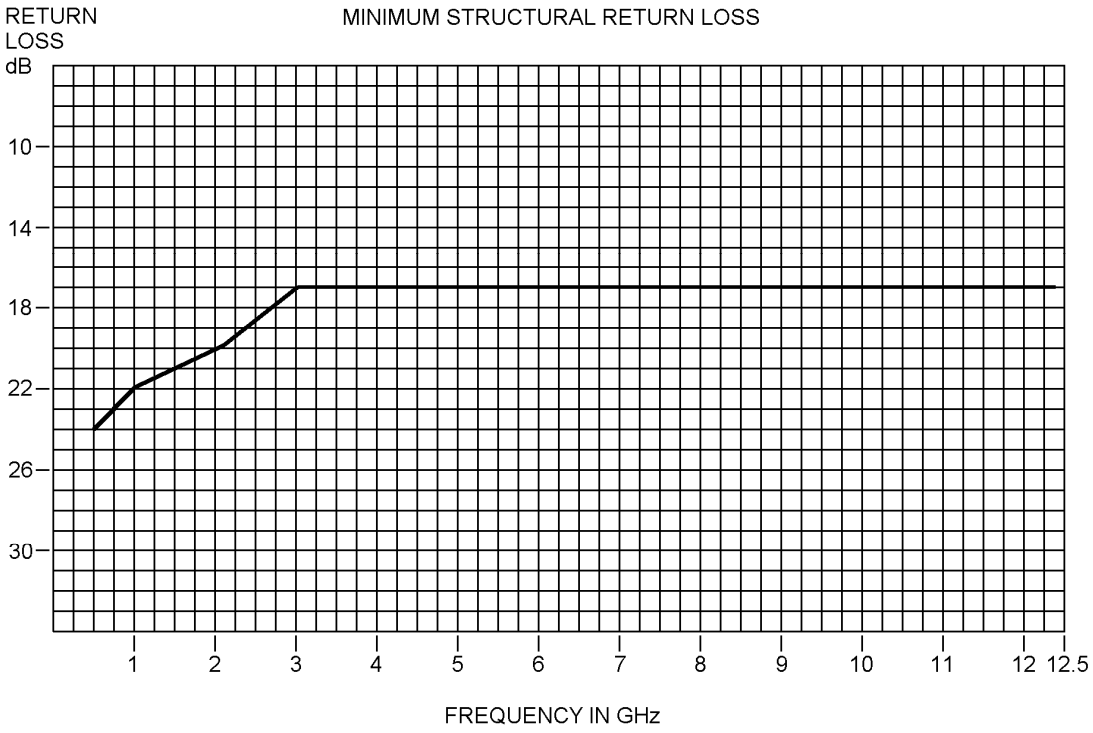
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Frequency MHz	Attenuation dB
50	3.2
100	4.5
400	10.5
1,000	17
3,000	38
10,000	78
12,400	90

FIGURE 2. Power rating and attenuation.

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TABULATED VALUES ARE FOR REFERENCE ONLY.
THE VALUES ON THE CHART REPRESENT THE REQUIREMENTS.

MHz	dB
50	25.5
400	23.8
1000	22
2000	20
3000	17
8000	17

SWR	RELECTION COEFFICIENT	RETURN LOSS Db
1.4985	.1995	14
1.4326	.1778	15
1.3767	.1585	16
1.3290	.1413	17
1.2880	.1259	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1726	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26

FIGURE 3. Structural return loss.

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Electrical:

Test frequency: 50 MHz to 12.4 GHz.

Spark test: 2,000 V rms, +10 percent, -0 percent.

Voltage withstanding: 3,000 V rms, +10, -0 percent.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 $\pm$ 2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Contamination: Not applicable.

Part or Identifying Number (PIN): M17/128-RG400.

The margins of this specification are marked with a vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-C-17, this document references the following:

MIL-PRF-39012

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

(Project 6145-2391-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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